

engineer, Delphi Electronics & Safety, shares, "The solid electrolyte interface and interfacial phenomena are the source of increase in resistance. You can monitor the ageing by monitoring the growth of the resistance with cycling. Since most battery pack manufacturers only look at the cell level as opposed to the component level, the resistance is a good enough parameter to follow."

Li-ion market drivers

Li-ion batteries are predominantly used in mobile phones, laptops, electric tools, cameras and many other handheld devices. But now these have started entering new applications like electric vehicles, plug-in hybrid vehicles, hybrid electric vehicles and electrical storage systems.

Li-ion batteries have a great potential to serve as a great-value alternate energy source, be it for powering the electric or hybrid vehicles or storing the energy recovered from industrial processes. Many electric vehicles and

hybrid vehicles plan to use Li-ion. However, it has not entered the mass production because of the high cost-volume ratio.

In India, LED lighting and UPS/inverter back-up applications are driving the growth of Li-ion batteries. Cellphones too are a huge market with constant demand for batteries.

In automobiles, large-format cells are preferred to small cells for the smaller footprint and lower component count. But larger cells also have larger heat dissipation. It is important to know how well you can actually assemble and produce the packs with smaller cells. For example, Panasonic is working with Tesla, the maker of high-performance electric vehicles, to develop its next generation of batteries to help Tesla lower the cost and improve the range of its vehicles.

Road ahead: Innovations to improve performance

Companies are making huge investments in R&D to improve Li-ion per-

formance in terms of capacity, safety, cycle times and life. Lots of new types of materials are being tested in combination to give better performance.

Toshiba plans to invest 25 billion yens in lithium-titanate batteries. The two major benefits of these batteries over other Li-ion batteries include no thermal runaway and higher cycle counts.

Panasonic has introduced an innovative nickel-oxide-based new platform technology that enables very high capacities to be achieved. In addition, the material and process technology prevent a negative alloy electrode from deforming with repeated charging.

Safety has been a major concern with batteries, especially in increased power state. "Patented heat-resistant layer technology now helps prevent overheating even in the case of an internal short-circuit. The solution consists of an insulated metal-oxide film between the electrodes," informs Shah. ■

The author is a senior technology journalist at EFY

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